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EXAMINER
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ART UNIT	PAPER NUMBER
3668	

NOTIFICATION DATE	DELIVERY MODE
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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/093,964	Applicant(s) Fang et al.	
	Examiner STEVEN V NGUYEN	Art Unit 3668	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 08 May 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-2,4-8 and 10-11 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,4-8 and 10-11 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date 05/08/2024, 05/08/2024.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

1. This office action is in response to the amendment filed on 05/08/2024. This action is made Final.

Notice of Pre-AIA or AIA Status

2. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Amendment

3. The amendment filed on 05/08/2024 has been entered. Claim 1-2, 4 – 8, 10 - 11 remain pending in the application. Applicant's amendment to the claims have overcome each and every objection previously set forth in the Non-Final Office Action mailed on 01/08/2024.

Response to Arguments

4. In the argument remarks, the Applicant argued that Kentley teaches generating the request and controlling the heating or cooling elements in response to the request, forming a dependency chain. For example, the cooling and heating elements are not controlled until the request has been received, and the request cannot be received until it has been generated. Therefore, Kentley fails to teach "provide an automated action plan in response to the automation description, wherein the automated action plan implements two branched operations of a plurality of automation commands, wherein the two branched operations have execution sequences independent of each other" (Remark, page 8 – 9)

The Examiner respectfully disagreed.

Kentley teaches receiving a temperature adjustment request from an external device to the vehicle (par. [0192]). Therefore, the request is not generated by the vehicle control system. Instead, the vehicle's control system interprets the request received from the external device and generates an

automated plan to execute the request accordingly. The temperature adjustment is implemented by heating and cooling elements, which are independent of each other. Thus, the heating element and cooling element operate as "two branched operations." It should also be obvious that the operation of the vehicle involves the operation of multiple components and systems to execute the action. For this reason, the Examiner believes that the teaching of Kentley still reads on the amendment.

Claim Rejections - 35 USC § 102

5. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –(a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

6. **Claims 1 –2, 4 - 5, 7 – 8, 10 - 11 are rejected under 35 U.S.C. 102(a)(1) as being anticipated by Kentley et al. (Publication No. US 20170126810 A1; hereafter Kentley)**

7. Regarding to claim 1, Kentley teaches **a system, comprising:**

a vehicle having a network comprising a plurality of end points *(see fig. 36A, plurality of "autonomous vehicle control system")*;

a controller *(see fig. 33 – 35, the "AV controller module")*, **comprising:**

an automation definition circuit structured to interpret an automation description generated by a device external to the vehicle, wherein the automation description is for at

least one of: a speaker system of the vehicle, a seat heating system of the vehicle, a seat positioning system of the vehicle, an environmental management system of the vehicle, a lightning system of the vehicle, or an off-vehicle messaging system of the vehicle; *([Par. 0151], "a user may control various features of the AV system **3602** from the user device **3600**, including an interior lighting control system **3604**, an ambient sound system **3606**, a temperature control system **3608**, a road handling preferences module **3610**, a rerouting request system **3616**, and a seating control system **3618**."; [Par. 0192], "An autonomous vehicle system comprising ambient features may receive **4100** a request from a user device to modify one of the ambient features through an autonomous vehicle service. For example, the request may be received **4100** at the autonomous vehicle system through the autonomous vehicle service over one or more networks and/or communication channels. A change parameter of the one of the ambient features may be determined **4102** from the request." Wherein the "user device" is external to the vehicle and allows the user to provide input for modifying operational conditions of the autonomous vehicle. Once the automation description has been received by the autonomous vehicle system, a change parameter can be determined.)*

an automation management circuit structured to:

provide an automated action plan in response to the automation description, wherein the automated action plan implements two branched operations of a plurality of automation commands, wherein the two branched operations have execution sequences independent of each other; *([Par. 0192], "the request may be received **4100** at the autonomous vehicle system through the autonomous vehicle service over one or more networks and/or communication channels. A change parameter of the one of the ambient features may be determined **4102** from the request. The change parameter may be determined **4102** through a format of the*

*request, such as the request being captured from the user device operating an application that communicates the request to the autonomous vehicle service. A change parameter may include a selection of a preconfigured mode for a sub-system, in one embodiment. In another embodiment, a change parameter may include a value of an adjustment to the ambient feature of the AV system."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein operations of the heating and cooling elements read on the "two branched operations")* **and**

store the automated action plan as a data file on a data storage communicatively coupled to the controller; *([Par. 0151], "a user may control various features of the AV system **3602** from the user device **3600**, including an interior lighting control system **3604**, an ambient sound system **3606**, a temperature control system **3608**, a road handling preferences module **3610**, a rerouting request system **3616**, and a seating control system **3618**. These preferences and/or configurations may be stored in association with the user and/or the user device **3600**"; [Par. 0150], "After the user is verified, the passenger security system **3650** may enable the user to enter the AV system **3602** by unlocking the door and/or opening the doors. In one embodiment after a user is verified, one or more stored configurations of parameters for features of the AV system **3602** may be automatically changed." This implies that the configuration changes requested by a specific user can be stored and subsequently applied the next time that particular user interacts with the vehicle.)*

an automation execution circuit structured to provide two automation command of the plurality of automation command in response to the automated action plan, wherein the two branched operations correspond to the two automation commands; *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein operations of the heating and cooling elements read on the "two branched operations" This should be understood that a command for changing the ambient features corresponding the change parameter is transmitted to a particular subsystem controller for execute the modification. The command for controlling each of the heating elements and cooling elements should be generated separately.)*

and

wherein an end point of the plurality of end points is responsive to at least one of the two branched automation commands to implement an automated vehicle response. *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein the*

"heating and cooling element" read on the "end point of the plurality of end points" and the command sends to each of the heating element and cooling element read on the two branched automation commands.)

Examiner Note: The components labeled as the 'automation definition circuit,' 'automation management circuit,' and 'automation execution circuit' are asserted to form integral parts of a unified controller. Kentley's system, as depicted in Figure 33, accomplishes the described functions through an 'AV controller module.' Consequently, it is to be inferred that the AV controller inherently encompasses all essential circuits required to execute the aforementioned functions.

8. Regarding to claim 2, Kentley teaches the system of claim 1.

Kentley further teaches **wherein the automated action plan implements a conditional operation of the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level." Where the "conditional operation" is to change the temperature to a certain level as requested by the user.)*

9. Regarding to claim 4, Kentley teaches the system of claim 1.

Kentley further teaches **wherein the automated action plan implements a state machine operation including the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the*

*temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein controlling the heating and cooling element to adjust the temperature to a certain temperature reads on " implementing a state machine")*

10. Regarding to claim 5, Kentley teaches the system of claim 1.

Kentley further teaches **wherein the automated action plan implements a cyclical operation including the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." This is interpreted as the heating and cooling element are controlled to regulate ambient temperature of the vehicle to a certain temperature requested by the user. And this process is repeated continuously to maintain the ambient temperature of the vehicle at a desired temperature.)*

11. Regarding to claim 7, Kentley teaches **A system, comprising:**

a vehicle having a network comprising a plurality of end points; *(see fig. 36A, plurality of "autonomous vehicle control system")***; and**

a controller *(see fig. 33 – 35, the "AV controller module")***, comprising:**

an automation definition circuit structured to interpret an automation description generated by a device external to the vehicle, wherein the automation description is not a request for a planned automated navigation route to a location and not a planned automated navigation route; *([Par. 0151], "a user may control various features of the AV system **3602** from the user device **3600**, including an interior lighting control system **3604**, an ambient sound system **3606**, a temperature control system **3608**, a road handling preferences module **3610**, a rerouting request system **3616**, and a seating control system **3618**."; [Par. 0192], "An autonomous vehicle system comprising ambient features may receive **4100** a request from a user device to modify one of the ambient features through an autonomous vehicle service. For example, the request may be received **4100** at the autonomous vehicle system through the autonomous vehicle service over one or more networks and/or communication channels. A change parameter of the one of the ambient features may be determined **4102** from the request." Wherein the "user device" is external to the vehicle and allows the user to provide input for modifying operational conditions of the autonomous vehicle. Once the automation description has been received by the autonomous vehicle system, a change parameter can be determined.)*

an automation management circuit structured to:

provide an automated action plan in response to the automation description, wherein the automated action plan implements two branched operations of a plurality of automation commands, wherein the two branched operations have execution sequences independent of each other; *([Par. 0192], "the request may be received **4100** at the autonomous vehicle system through the autonomous vehicle service over one or more networks and/or communication channels. A change*

*parameter of the one of the ambient features may be determined **4102** from the request. The change parameter may be determined **4102** through a format of the request, such as the request being captured from the user device operating an application that communicates the request to the autonomous vehicle service. A change parameter may include a selection of a preconfigured mode for a sub-system, in one embodiment. In another embodiment, a change parameter may include a value of an adjustment to the ambient feature of the AV system."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein operations of the heating and cooling elements read on the "two branched operations")),* **and**

store the automated action plan as a data file on a data storage communicatively coupled to the controller; *([Par. 0151], "a user may control various features of the AV system **3602** from the user device **3600**, including an interior lighting control system **3604**, an ambient sound system **3606**, a temperature control system **3608**, a road handling preferences module **3610**, a rerouting request system **3616**, and a seating control system **3618**. These preferences and/or configurations may be stored in association with the user and/or the user device **3600**"; [Par. 0150], "After the user is verified, the passenger security system **3650** may enable the user to enter the AV system **3602** by unlocking the door and/or opening the doors. In one embodiment after a user is verified, one or more stored configurations of parameters for features of the AV system **3602** may be automatically changed." This implies that the configuration changes*

requested by a specific user can be stored and subsequently applied the next time that particular user interacts with the vehicle.) **and**

an automation execution circuit structured to provide two automation commands of the plurality of automation commands in response to the automated action plan, wherein the two branched operations correspond to the two automation commands; *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein operations of the heating and cooling elements read on the "two branched operations". This should be understood that a command for changing the ambient features corresponding the change parameter is transmitted to a particular subsystem controller for execute the modification. The command for controlling each of the heating elements and cooling elements should be generated separately.)* **and**

wherein an end point of the plurality of end points is responsive to at least one of the two branched automation commands to implement an automated vehicle response. *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a*

*heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein the "heating and cooling element" read on the "end point of the plurality of end points" and the command sends to each of the heating element and cooling element read on the two branched automation commands.)*

Examiner Note: The components labeled as the 'automation definition circuit,' 'automation management circuit,' and 'automation execution circuit' are asserted to form integral parts of a unified controller. Kentley's system, as depicted in Figure 33, accomplishes the described functions through an 'AV controller module.' Consequently, it is to be inferred that the AV controller inherently encompasses all essential circuits required to execute the aforementioned functions.

12. Regarding to claim 8, Kentley teaches the system of claim 7.

Kentley further teaches **wherein the automated action plan implements a conditional operation of the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level." Where the "conditional operation" is to change the temperature to a certain level as requested by the user.)*

13. Regarding to claim 10, Kentley teaches the system of claim 7.

Kentley further teaches **wherein the automated action plan implements a state machine operation including the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the*

*temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." Wherein controlling the heating and cooling element to adjust the temperature to a certain temperature reads on " implementing a state machine")*

14. Regarding to claim 11, Kentley teaches the system of claim 7.

Kentley further teaches **wherein the automated action plan implements a cyclical operation including the automation command.** *([Par. 0192], "The change parameter of the one of the ambient features may then be executed **4104** at the autonomous vehicle system. For example, the temperature may be changed by the temperature control system of the autonomous vehicle responsive to a request to change the temperature to a certain level."; [Par. 0153], "A temperature control system **3608** may control an interior temperature of the AV system **3602** through heating and cooling elements, such as a heater and an air conditioning unit, of the AV system **3602**. In one embodiment, the user may set a desired temperature for the AV system **3602** through the user device **3600**." This is interpreted as the heating and cooling element are controlled to regulate ambient temperature of the vehicle to a certain temperature requested by the user. And this process is repeated continuously to maintain the ambient temperature of the vehicle at a desired temperature.)*

Claim Rejections - 35 USC § 103

15. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

16. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

17. **Claim 6 is rejected under 35 U.S.C. 103 as being unpatentable over Kentley in view of Nakashima et al. (Publication No. US 20210024035 A1; hereafter Nakashima).**

18. Regarding to claim 6, Kentley teaches the system of claim 1.

Kentley teaches to adjust operational condition of the autonomous vehicle based on input from an external device as described in claim 1 above, but does not explicitly disclose **wherein the automation description is at least in part for a door locking system of the vehicle.**

However, Nakashima teaches **wherein the automation description is at least in part for a door locking system of the vehicle.** *([Par. 0091], "The user terminal **37** is preferably a mobile terminal such as a smartphone, a mobile phone, or a tablet terminal, and at least needs to be portable to be also used as a vehicle door lock key as in the second embodiment described later."; [Par. 0087], "By controlling the door lock actuator **29**, the locking and unlocking of the door lock of the vehicle can be remotely controlled via the server **3** based on the locking and unlocking commands from the user terminal.")*

It would have been obvious to one of ordinary skill in the art before the effective filing date of the claim invention to modify Kentley to incorporate the teaching of Nakashima. The modification is apparent as it involves incorporating an automation description for door locking and unlocking in the vehicle. This enhancement enables users to conveniently control the door lock/unlock functionality through their personal devices.

Conclusion

19. **THIS ACTION IS MADE FINAL.** Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

20. Any inquiry concerning this communication or earlier communications from the examiner should be directed to STEVEN V NGUYEN whose telephone number is (571)272-7320. The examiner can normally be reached Monday -Friday 11am - 7pm EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, James J Lee can be reached on (571) 270-5965. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/STEVEN VU NGUYEN/
Examiner, Art Unit 3668

/JAMES J LEE/
Supervisory Patent Examiner, Art Unit 3668